POWER BI PORTFOLIO PROJECT

# Pizza Sales Analytics

## Performance & Business Insights Dashboard

*An end-to-end business intelligence solution that transforms 48,620 sales records into actionable insights across revenue, product performance, customer preferences, and demand patterns.*

**Power BI Desktop** **Power Query (M)** **DAX**

Jan–Dec 2015 | 4 categories · 5 sizes · 91 menu items



BUSINESS CONTEXT

# Business Problem & Objectives

## THE CHALLENGE

The business had a full year of transaction-level pizza sales data, but no consolidated performance view. Understanding revenue drivers, product performance, and peak demand required manual analysis.

## SUCCESS CRITERIA

- Create a **single source of truth** for core sales KPIs
- Identify **high- and low-performing products**
- Surface **peak and low-demand periods**

## KEY BUSINESS QUESTIONS ANSWERED

1. How is the business performing across **revenue, orders, and order value?**
2. Which pizzas are the **strongest and weakest revenue contributors?**
3. How do sales change across **months, quarters, days, and hours?**
4. When do **peak sales periods** occur?
5. What practical actions can improve results without disrupting operations?

DATA FOUNDATION

# Dataset Overview

| 48,620 | 2015 | 4 | 5 | 91 |
|---|---|---|---|---|
| Transaction Records | Full-Year Data | Pizza Categories | Pizza Sizes | Menu Items |

## KEY FIELDS

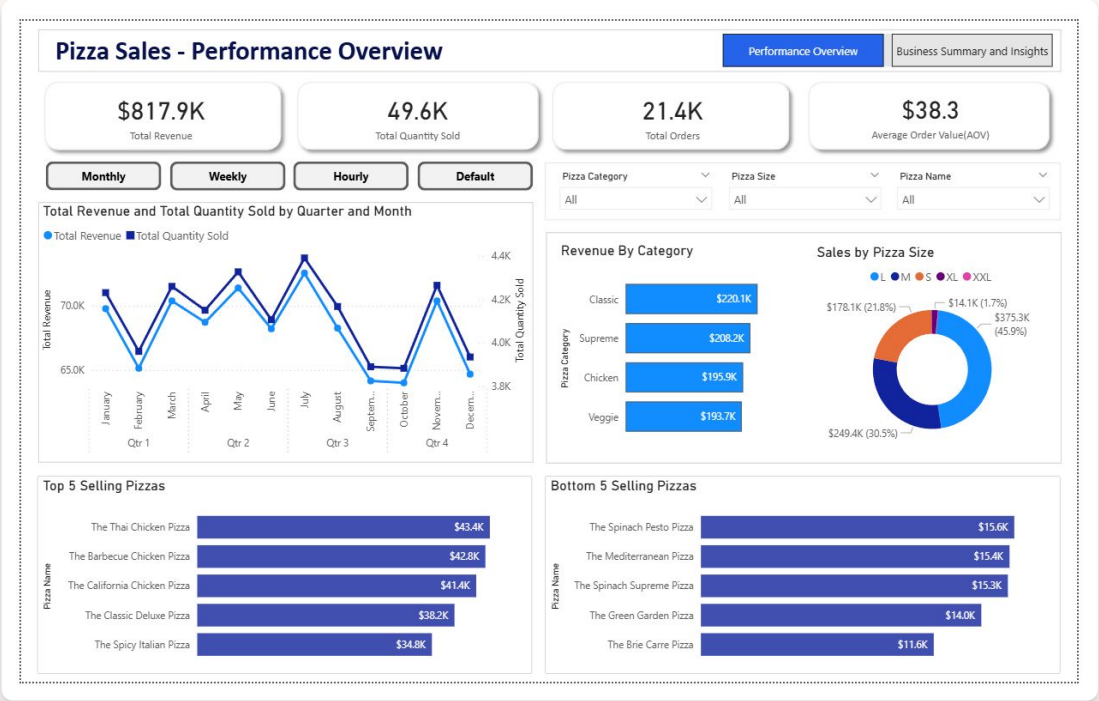
- Order ID, Order Date, Order Time
- Pizza Name, Category & Size
- Quantity, Unit Price, Total Price

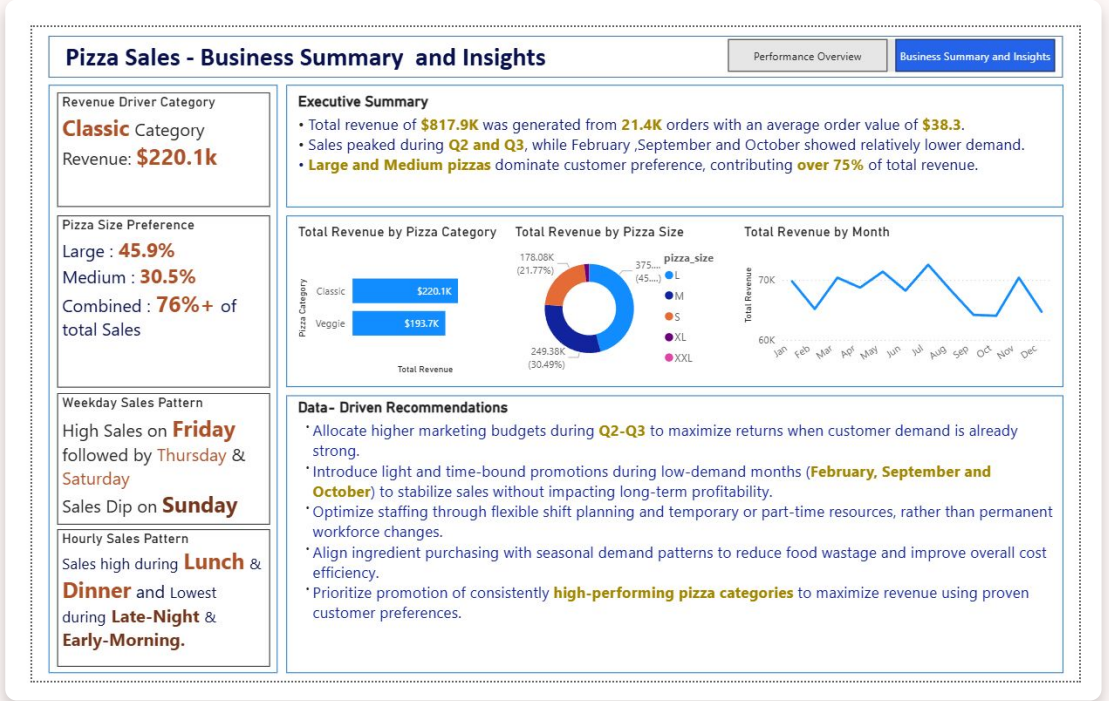
## DATA PREPARATION & MODELING

- Validated data types across date, time, numeric & text fields
- Checked for nulls and standardized pizza names, categories & sizes
- Engineered Order Day, Month Name/Number, Quarter & Hour of Order for time-based analysis

THE DELIVERABLE

# Two Interactive Dashboard Pages




## Performance Overview

**KPI monitoring, revenue trends, product performance and sales patterns**

## Business Summary & Insights

**Key findings and data-driven recommendations for decision-making**

**KPI Monitoring**
Revenue · Orders · AOV · Quantity Sold

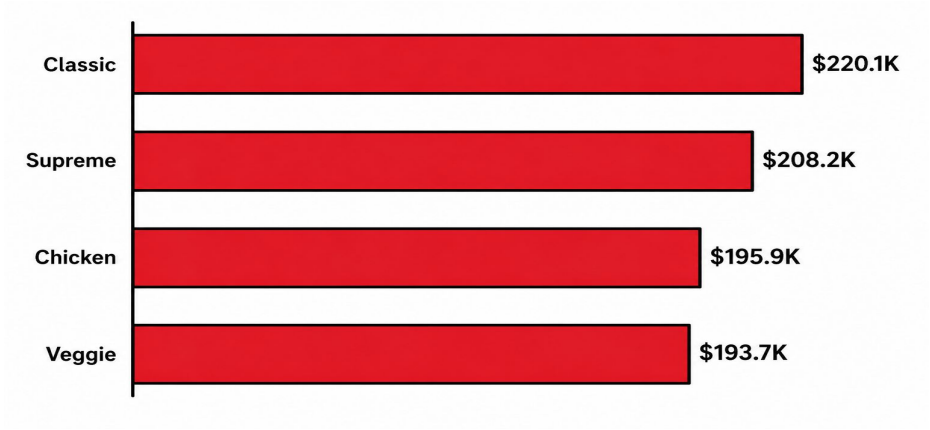
**Cross-Filtering**
Category · Size · Pizza Name

**Time Analysis**
Monthly · Weekly · Hourly

WHAT THE DATA SHOWS

# Key Business Insights

### Revenue by Pizza Category




*Category performance is relatively balanced — Classic leads at $220.1K, while the four categories remain within ~12% of each other. Product and size offer stronger differentiation.*

1

### Seasonal Rhythm

Sales peak during **Q2–Q3**, with **July recording the strongest monthly performance**. Demand is comparatively lower in **February, September, and October**.

2

### Size Preference

**Large and Medium pizzas account for 76%+ of total revenue**, contributing approximately **$375.3K and $249.4K**, respectively.

3

### Weekly Pattern

**Friday is the strongest sales day**, followed by Thursday and Saturday, while **Sunday shows a noticeable dip**.

4

### Daypart Pattern

Demand is concentrated around **lunch and dinner**, while **late-night and early-morning periods remain comparatively quiet**.

TURNING INSIGHT INTO ACTION

# Business Recommendations

## Marketing

- Double down on proven winners: **Thai Chicken, Barbecue Chicken, Classic Deluxe**
- Use **targeted, time-bound promotions** in Feb, Sep & Oct to address seasonal dip

## Inventory

- Prioritize stock for **Large & Medium sizes** and the **Classic/Supreme categories**
- Align ingredient purchasing with seasonal demand to **reduce waste and stock risk**

## Staffing & Operations

- Use flexible staffing around **Thu-Sat and lunch–dinner peaks**
- Optimize shift allocation rather than increasing permanent headcount

## Product Strategy

- Introduce combo offers to lift Average Order Value beyond $38.3
- **Reposition or test alternatives** for persistent bottom performers such as **Brie Carre & Green Garden**

**Expected Business Impact** ↑ Revenue | ↓ Waste | ↑ AOV | Better Labor Utilization

BEHIND THE DASHBOARD

# Technical Implementation

1

### Data Loading

Imported CSV data into Power BI and validated data types across **dates, times, numeric and text fields**.

2

### Data Cleaning

Performed **null checks, consistency validation and standardization** across pizza names, categories, sizes and prices.

3

### Feature Engineering

Created Power Query features including **Order Day, Month, Month Number, Quarter and Order Hour** to support time-based analysis.

4

### DAX Measures

Built reusable KPI measures for **Total Revenue, Total Orders, Quantity Sold and Average Order Value**.

5

### Visualization & UX

Designed a **two-page interactive dashboard** with custom slicer buttons, cross-filtering and dynamic **Monthly / Weekly / Hourly** analysis.

*Custom slicer buttons let users pivot between Monthly, Weekly and Hourly views on a single canvas — an interaction pattern chosen for usability over building three separate report pages.*

BUSINESS IMPACT

# Business Value & Skills Demonstrated

## VALUE DELIVERED

- Replaces manual spreadsheet analysis with an **interactive, filterable view** of revenue, orders and average order value.
- Identifies which **pizzas, sizes and categories** deserve promotion, inventory priority or repositioning.
- Surfaces **day-of-week, hourly and seasonal demand patterns** to support staffing and inventory planning.
- Connects analytical findings to **practical business actions** within real operational constraints.

## SKILLS DEMONSTRATED

| | |
|---|---|
| **Power BI** | **DAX** |
| **Power Query (M)** | **Data Preparation** |
| **Data Visualization** | **Business Analysis** |
| **Dashboard UX Design** | **Data Storytelling** |

*Full interactive dashboard, PBIX file and case study available on request.*